

Systemic Lupus Erythematosus

Addis Ababa University
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Definition

Chronic, multisystem autoimmune disease

Characterized by:

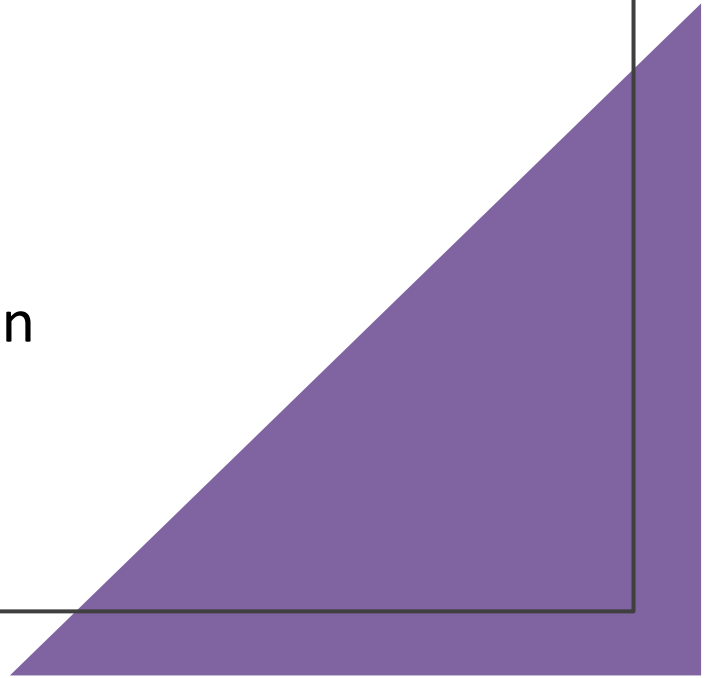
- Loss of immune tolerance
- Autoantibody production (anti-dsDNA, anti-Sm)
- Immune complex deposition → inflammation and organ damage
- Predominantly affects women of reproductive age

Epidemiology

- Global Prevalence and Incidence
 - Prevalence: Approximately 20–150 cases per 100,000 population worldwide
 - Incidence: Around 1–10 new cases per 100,000 person-years
 - Wide geographic variation due to:
 - Genetic susceptibility
 - Environmental exposures
 - Differences in case ascertainment and healthcare access

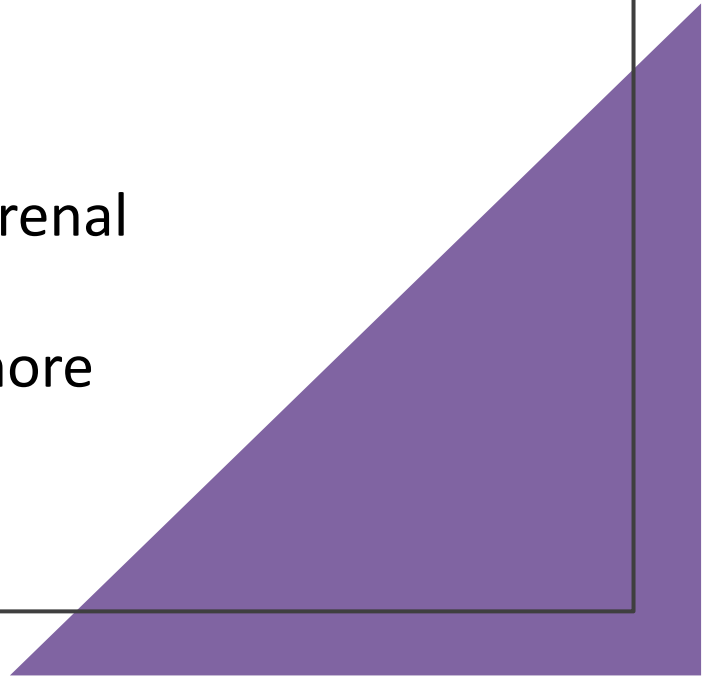
Epidemiology ...

- Sex Distribution
 - Strong female predominance
 - Female-to-male ratio: ~9–10 : 1
 - Sex disparity is greatest during reproductive age, highlighting hormonal influences (estrogen effects on immunity)



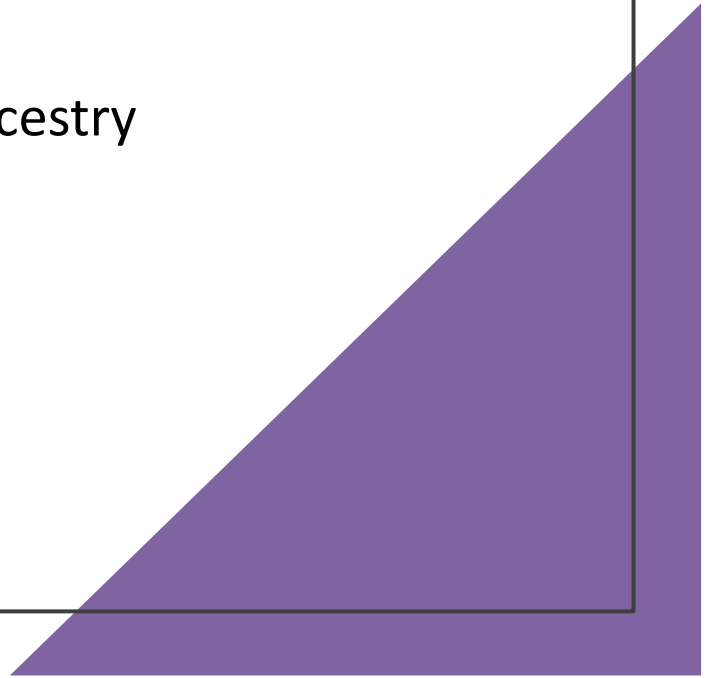
Epidemiology ...

- Age Distribution
 - Typical onset between 15–45 years
 - Can occur at any age:
 - Childhood-onset SLE: Often more severe, higher renal and CNS involvement
 - Late-onset SLE (>50 years): Less aggressive but more comorbidities



Epidemiology ...

- Ethnic and Racial Variations
 - Higher prevalence and severity among:
 - People of African, Asian, Hispanic, and Indigenous ancestry
 - These groups tend to have:
 - Earlier disease onset
 - Higher rates of lupus nephritis
 - Increased disease activity and damage



Epidemiology ...

- Geographic Patterns
 - Higher prevalence reported in:
 - North America
 - Latin America
 - Parts of Asia
 - Lower reported prevalence in some African regions, likely reflecting under diagnosis rather than true absence of disease



Epidemiology

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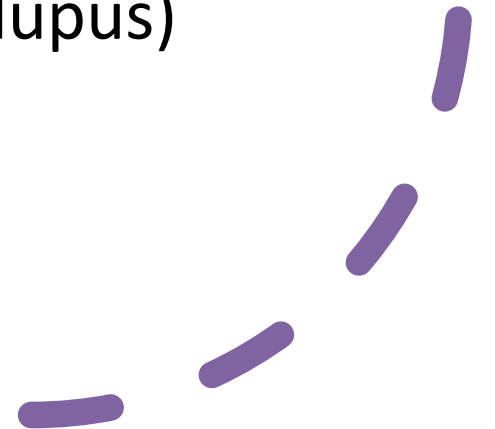
- Genetic Epidemiology
 - Strong genetic contribution:
 - First-degree relatives have a 10–20-fold increased risk
 - Higher concordance in monozygotic twins
 - Associated with multiple immune-regulatory genes (e.g., HLA, complement pathways)



Epidemiology

...

- Environmental and Hormonal Risk Factors
 - Ultraviolet (UV) light exposure
 - Infections (viral triggers)
 - Smoking
 - Hormonal factors (estrogen)
 - Certain drugs (drug-induced lupus)



Epidemiology

...

- Mortality and Survival Trends
 - Marked improvement in survival over recent decades
 - 10-year survival: >90% in many settings
 - Leading causes of death:
 - Infections
 - Cardiovascular disease
 - Renal failure
 - Neuropsychiatric complications



Epidemiology

...

- Disease Burden and Public Health Impact
 - Chronic relapsing course
 - Significant impact on:
 - Quality of life
 - Productivity
 - Healthcare utilization
 - Disproportionate burden in women of childbearing age



Pathophysiology

- Key Pathogenic Mechanisms
 - Loss of immune tolerance → autoantibody production (Anti-Double-Stranded DNA (anti-dsDNA), Anti-Smith antibodies (anti-Sm))
 - Immune complex deposition → complement activation
 - Dysregulated B cells, T cells, and innate immunity
 - Overactivation of Type I interferon (IFN) pathway
 - Increased BAFF (BLyS) promoting B-cell survival



Drug-Induced Lupus



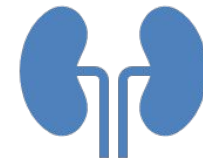
Common causative drugs

Procainamide, hydralazine (highest risk), isoniazid, minocycline, methyldopa, anti-TNF agents, interferon- α , sulfasalazine, carbamazepine.



Clinical features

Arthralgia, myalgia, fever, rash, and serositis; renal and CNS involvement are rare.



Laboratory findings

Positive ANA, anti-histone antibodies common; anti-dsDNA usually absent (except with anti-TNF agents); complement levels typically normal.

Drug-Induced Lupus

Onset

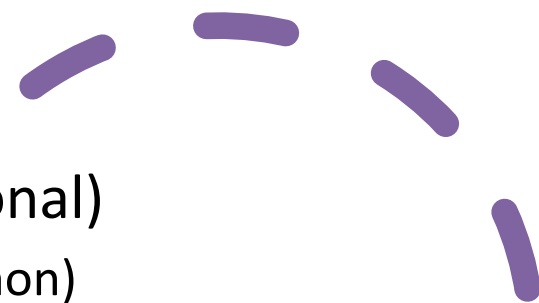
- Months to years after drug exposure.

Management:

- Discontinuation of the offending drug is key;
- symptoms usually resolve within weeks to months,
- with NSAIDs or short-term steroids if needed.



Clinical Pictures



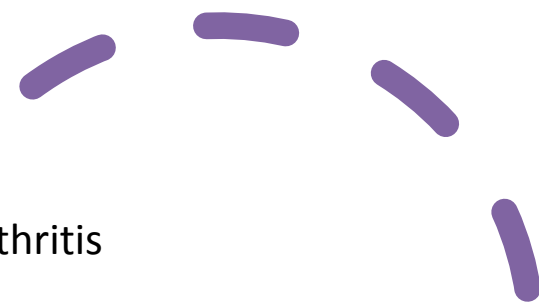
1. General (Constitutional)

- Fatigue (most common)
- Fever (often low-grade, non-infectious)
- Weight loss
- Malaise
- Night sweats



Clinical Pictures...

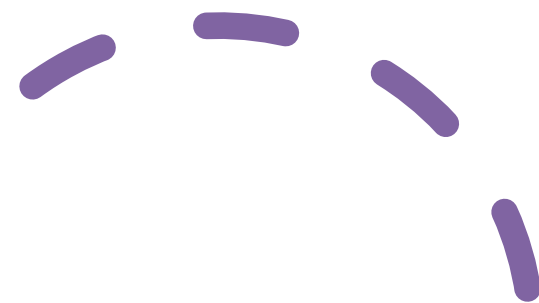
2. Musculoskeletal Manifestations

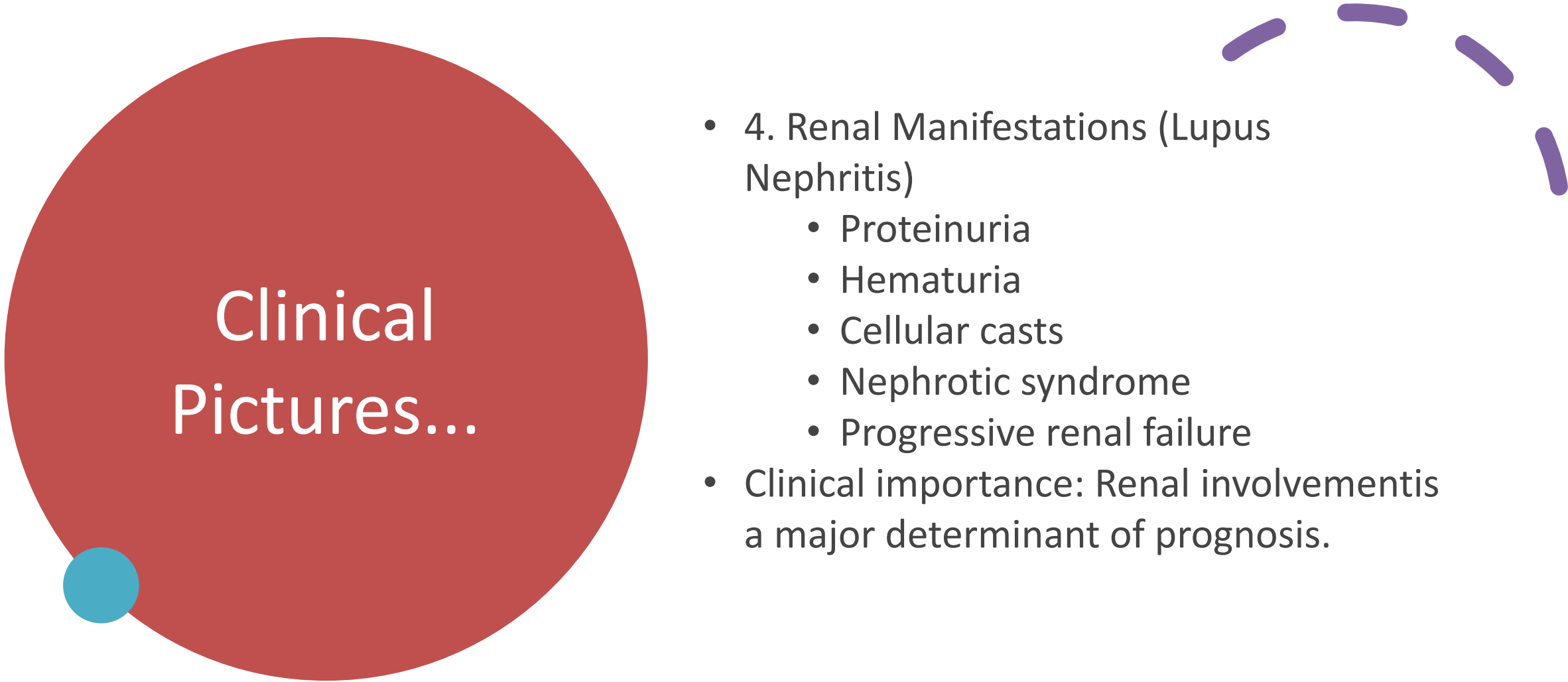
- Symmetrical inflammatory polyarthritis
 - Arthralgia and myalgia
 - Non-erosive arthritis (Jaccoudarthropathy in chronic disease)
 - Tendonitis
- 



Clinical Pictures...

3. Cutaneous and Mucosal Manifestations

- Malar (butterfly) rash
 - Discoid rash
 - Photosensitivity
 - Oral and nasal ulcers (usually painless)
 - Alopecia (non-scarring or scarring)
 - Livedo reticularis
 - Raynaud phenomenon
- 



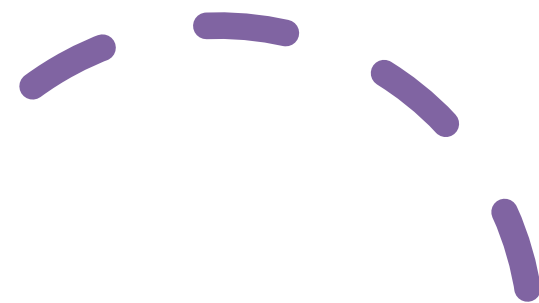
Clinical Pictures...

- 4. Renal Manifestations (Lupus Nephritis)
 - Proteinuria
 - Hematuria
 - Cellular casts
 - Nephrotic syndrome
 - Progressive renal failure
- Clinical importance: Renal involvement is a major determinant of prognosis.



Clinical
Pictures...

5. Neuropsychiatric Manifestations

- § Headache
 - § Cognitive dysfunction
 - § Seizures
 - § Psychosis
 - § Stroke
 - § Peripheral neuropathy
- 



Clinical Pictures...

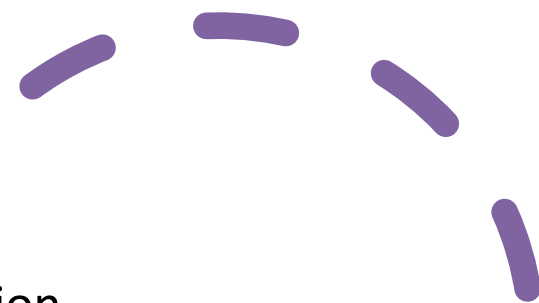


6. Cardiovascular Manifestations

- Pericarditis (most common cardiac manifestation)
- Myocarditis
- Accelerated atherosclerosis
- Libman–Sacks endocarditis
- Increased thromboembolic events (especially with antiphospholipid syndrome)



Clinical Pictures...



7. Pulmonary Manifestations

- Pleuritis and pleural effusion
- Interstitial lung disease
- Pulmonary hypertension
- Acute lupus pneumonitis
- Shrinking lung syndrome (rare)



Clinical Pictures...

- **8. Hematologic Manifestations**

- Anemia (chronic disease or hemolytic)
- Leukopenia
- Lymphopenia
- Thrombocytopenia
- Antiphospholipid syndrome–related thrombosis



Clinical Pictures...



9. Gastrointestinal and Hepatic Manifestations

- § Nonspecific abdominal pain
- § Mesenteric vasculitis
- § Lupus enteritis
- § Hepatitis (rare)
- § Pancreatitis



Clinical Pictures...



10. Reproductive and Obstetric Manifestations

- § Recurrent pregnancy loss
- § Pre-eclampsia
- § Intrauterine growth restriction
- § Neonatal lupus (anti-Ro/La antibodies)

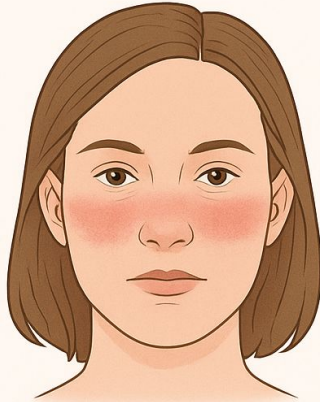


Clinical Pictures...

11. Disease Course Patterns

- Relapsing–remitting (most common)
 - Chronic active disease
 - Monophasic (rare)
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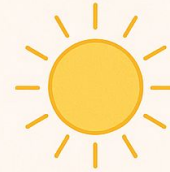
HALLMARK CLINICAL FEATURES OF SLE



Malar (butterfly) rash
over cheeks and bridge
of nose



Discoid rash
raised scaly
patches



Photosensitivity
rash triggered
by sunlight



**Oral/nasal
ulcers**
often painless



**Arthralgia /
arthritis**
Non-erosive
inflammation
Typically
symmetric



Myalgia
muscle aches



**Hematologic
Abnormalities**
Anemia
Leukopenia
Lymphopenia

Red Flags Suggestive of SLE

Young woman with multisystem involvement

Photosensitive rash + arthritis + cytopenia

Renal abnormalities with positive ANA

Unexplained fever and fatigue



EULAR → *European Alliance of Associations for Rheumatology* (formerly European League Against Rheumatism).

ACR → *American College of Rheumatology*.

Diagnosis of SLE



2023 EULAR/ACR criteria



Autoantibodies: Antinuclear antibodies (ANA), anti-dsDNA, anti-Sm



Complement levels

Goal of Therapy

Control

Control disease activity

Prevent

Prevent flares

Minimize

Minimize organ damage

Reduce

Reduce corticosteroid exposure

Improve

Improve quality of life

Goal of Therapy...

Treat to target: remission or low disease activity (LLDAS)

Individualized therapy based on:

- Organ involvement
- Disease severity
- Comorbidities
- Pregnancy status

Management

Induction vs maintenance

Hydroxychloroquine
cornerstone

Biologics: belimumab,
anifrolumab

Glucocorticoid minimization

Core Drug Classes in SLE

- **Antimalarials (Cornerstone Therapy)**
 - Hydroxychloroquine (HCQ)
 - Recommended for ALL SLE patients unless contraindicated
 - Inhibits Toll-like receptor (TLR) signaling
 - Reduces antigen presentation
 - Decreases cytokine production

- Clinical Benefits
 - ↓ disease flares
 - ↓ thrombotic risk
 - ↓ mortality
 - Improves lipid profile & glucose control
- Key Safety Points
 - Retinal toxicity (dose & duration dependent)
 - Annual ophthalmologic screening after 5 years (earlier if high risk)

Evidence Note: Strong mortality benefit; withdrawal increases flare risk.

Glucocorticoids

- Role
 - Rapid control of inflammation
 - Induction therapy for moderate–severe disease
- Forms
 - Oral prednisone/prednisolone
 - IV methylprednisolone pulses (severe organ involvement)
- Current Best Practice
 - Lowest effective dose
 - Short duration
 - Early steroid-sparing agents
- Adverse Effects
 - Infection
 - Osteoporosis
 - Hypertension
 - Diabetes
 - Cardiovascular risk

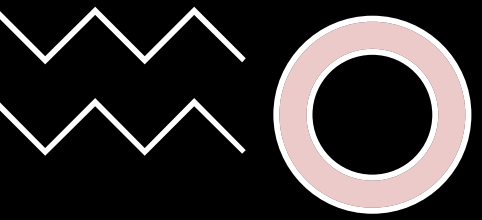
Target: ≤ 5 mg/day prednisone equivalent for maintenance (if possible)

Conventional Immunosuppressive Agents

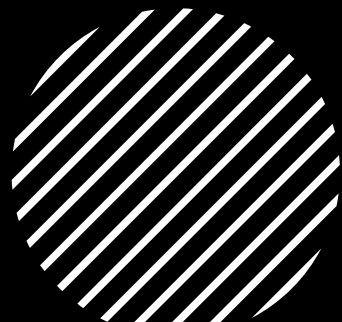
- **Methotrexate**
 - Mild–moderate disease (arthritis, skin)
 - Steroid-sparing
- **Azathioprine**
 - Maintenance therapy
 - Safe in pregnancy (preferred immunosuppressant)
- **Mycophenolate Mofetil (MMF)**
 - First-line for:
 - Lupus nephritis (Class III–V)
 - Severe systemic disease
 - Better safety profile than cyclophosphamide
- **Cyclophosphamide**
 - Severe, life-threatening disease
 - Neuropsychiatric lupus
 - Severe nephritis
 - IV low-dose regimens preferred (toxicity reduction)

Biologic and Targeted Therapies (Modern SLE Management)

- **Belimumab (Anti-BLyS/BAFF)**
 - **Indications**
 - Active, autoantibody-positive SLE
 - Add-on to standard therapy
 - Approved for lupus nephritis
 - **Benefits**
 - ↓ flares
 - ↓ steroid requirement
- **Anifrolumab (Anti-Type I Interferon Receptor)**
 - Blocks IFN- α signaling (key SLE pathway)
 - Moderate–severe SLE (non-renal, non-CNS)
 - Improves skin and joint disease
 - Steroid-sparing effect
- **Rituximab (Anti-CD20)**
 - Off-label but widely used
 - Refractory SLE
 - Severe hematologic or renal disease



Lupus Nephritis: Pharmacologic Approach



Induction

- MMF + glucocorticoids
OR
- Low-dose cyclophosphamide + glucocorticoids

Maintenance

- MMF or azathioprine
- HCQ continued in all patients

Emerging Add-Ons

- Belimumab
- Calcineurin inhibitors (e.g., voclosporin)

Adjunctive & Supportive Pharmacotherapy

ACE inhibitors / ARBs → proteinuria

Statins → cardiovascular risk

Calcium + Vitamin D → bone protection

Anticoagulation → antiphospholipid syndrome

Vaccinations (non-live preferred)

SLE in Special Populations Pregnancy

- Safe:
 - HCQ
 - Azathioprine
 - Low-dose steroids
- Contraindicated:
 - MMF
 - Cyclophosphamide
 - Methotrexate

- **Infection Risk**
 - Immunosuppression
↑ risk
 - Screen for TB,
hepatitis before
biologics

Monitoring

Disease activity
(SLEDAI)

Renal monitoring

Infection surveillance

Key Take-Home Messages

HCQ is foundational for all SLE patients

Steroid minimization is a major treatment goal

Biologics are transforming SLE outcomes

Treatment must be individualized and organ-specific

Long-term monitoring is essential to prevent damage

References

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2023 EULAR recommendations for the management of systemic lupus erythematosus.
Annals of the Rheumatic Diseases. 2023.
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2019 EULAR/ACR classification criteria for systemic lupus erythematosus.
Annals of the Rheumatic Diseases. 2019.
- **Hahn BH, et al.**
American College of Rheumatology guidelines for screening, treatment, and management of lupus nephritis.
Arthritis Care & Research. 2012 (still foundational, with updates integrated into EULAR guidance).

